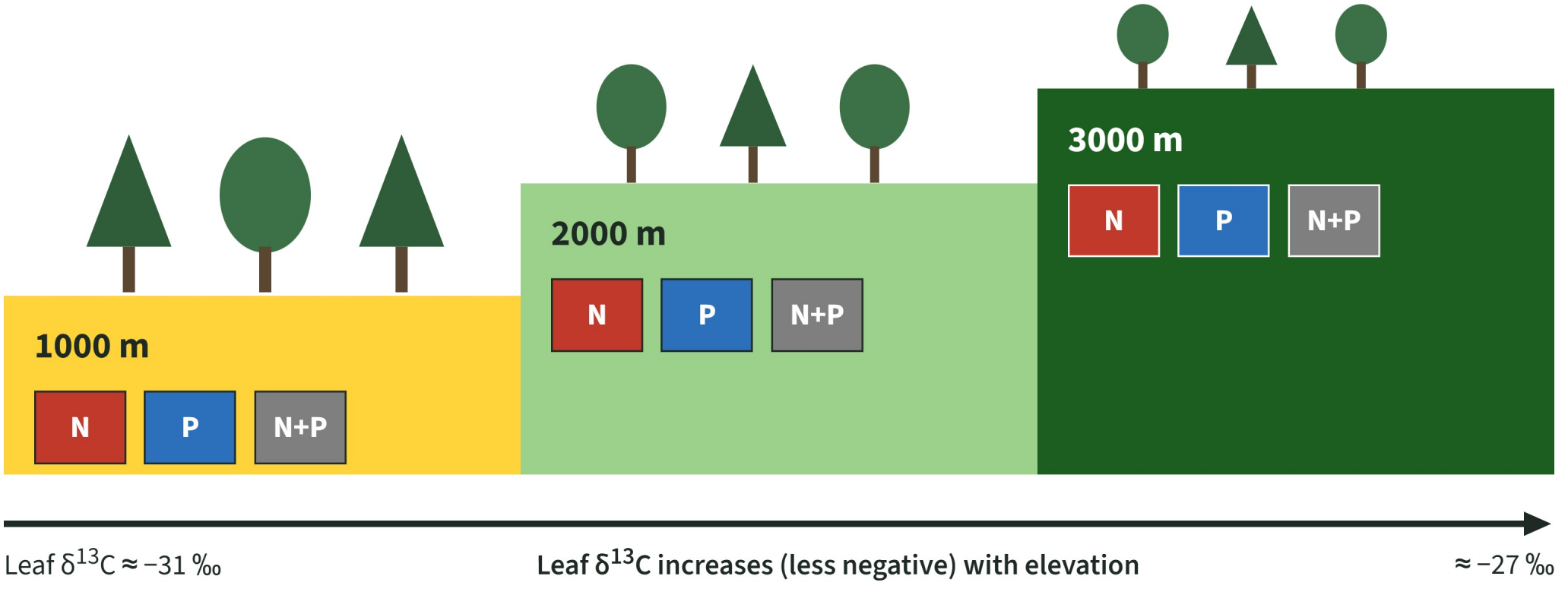
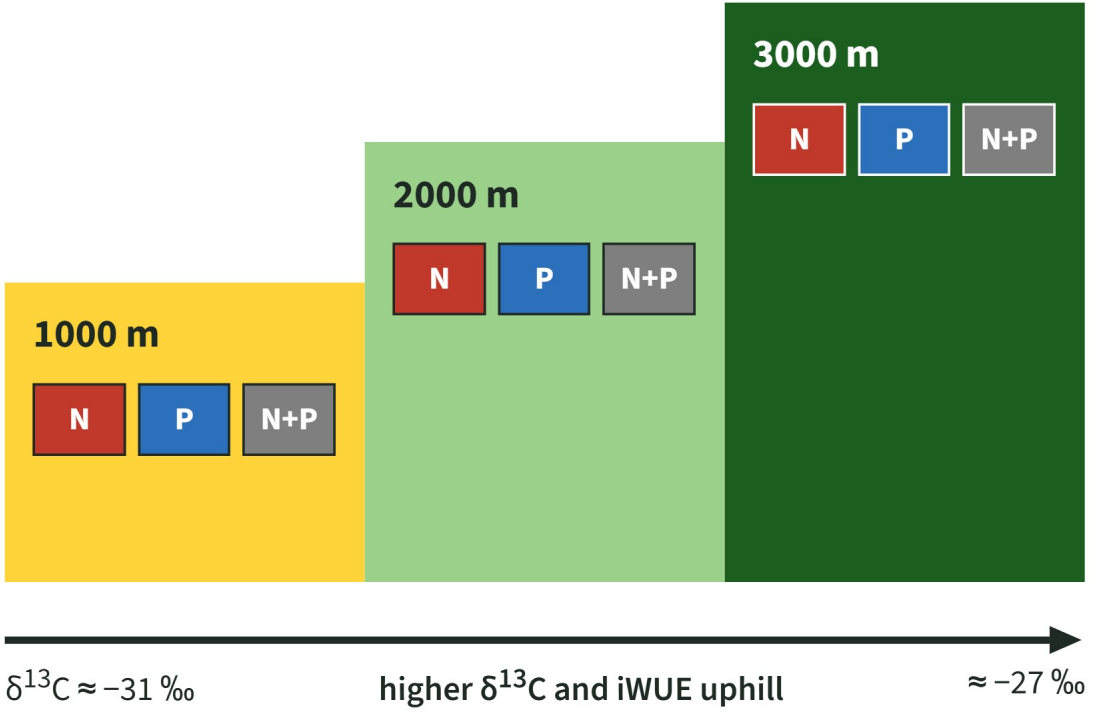
18 years of N and P addition along an Andean elevation gradient

NUMEX experiment, tropical montane forest, southern Ecuador



More leaf nutrients, same water-use efficiency

18 years of N and P addition, NUMEX, southern Ecuador



Leaf nutrients increased

Leaf P rose in 5 of 6 species (up to +272 %); leaf N rose by 11–20 % in 4 species.

iWUE and $\delta^{13}C$ did not change

No consistent response to N, P or N+P in any species; differences were set by site and species.

iWUE tracks leaf structure

Within species, iWUE was higher in denser leaves (lower SLA), not in leaves with more N.